

Slides: "7. Data Modelling
Using ER Model 1".

LECTURE-19

Monday.
13/4/20

Entity-Relationship Model:

→ Specific objects with an independent existence.

→ Attributes: Properties used to describe an entity.

→ Simple: has single atomic value e.g.: SSN

→ Composite: Consists of several components e.g.: Address, Fullname

→ Multi-valued: Attribute have multiple values for that e.g.: prev. degree^{of student} or colors in car.

→ this attribute will be double circled

* NOTE: ER model is first step we do after taking info about DB from clients and at this moment, no tables or anything exist.

→ Relationships: Relates two or more distinct entities with a specific meaning. e.g.: EMPLOYEE works on PROJECTS.

→ Cardinality: (1:1, 1:N, N:M) → (one-to-one, one-to-many, many-to-many).

e.g.: 1 Employee can work for multiple project (1:N) N will come with projects.

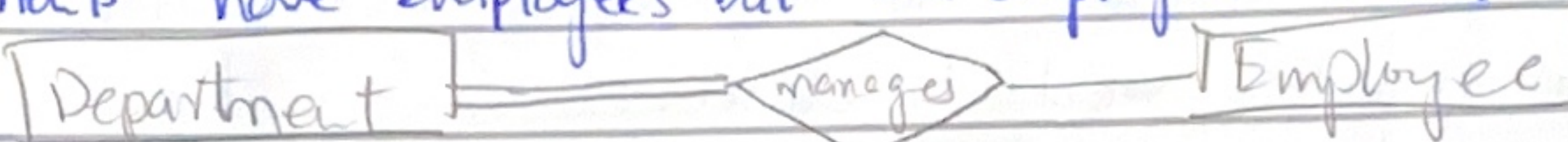
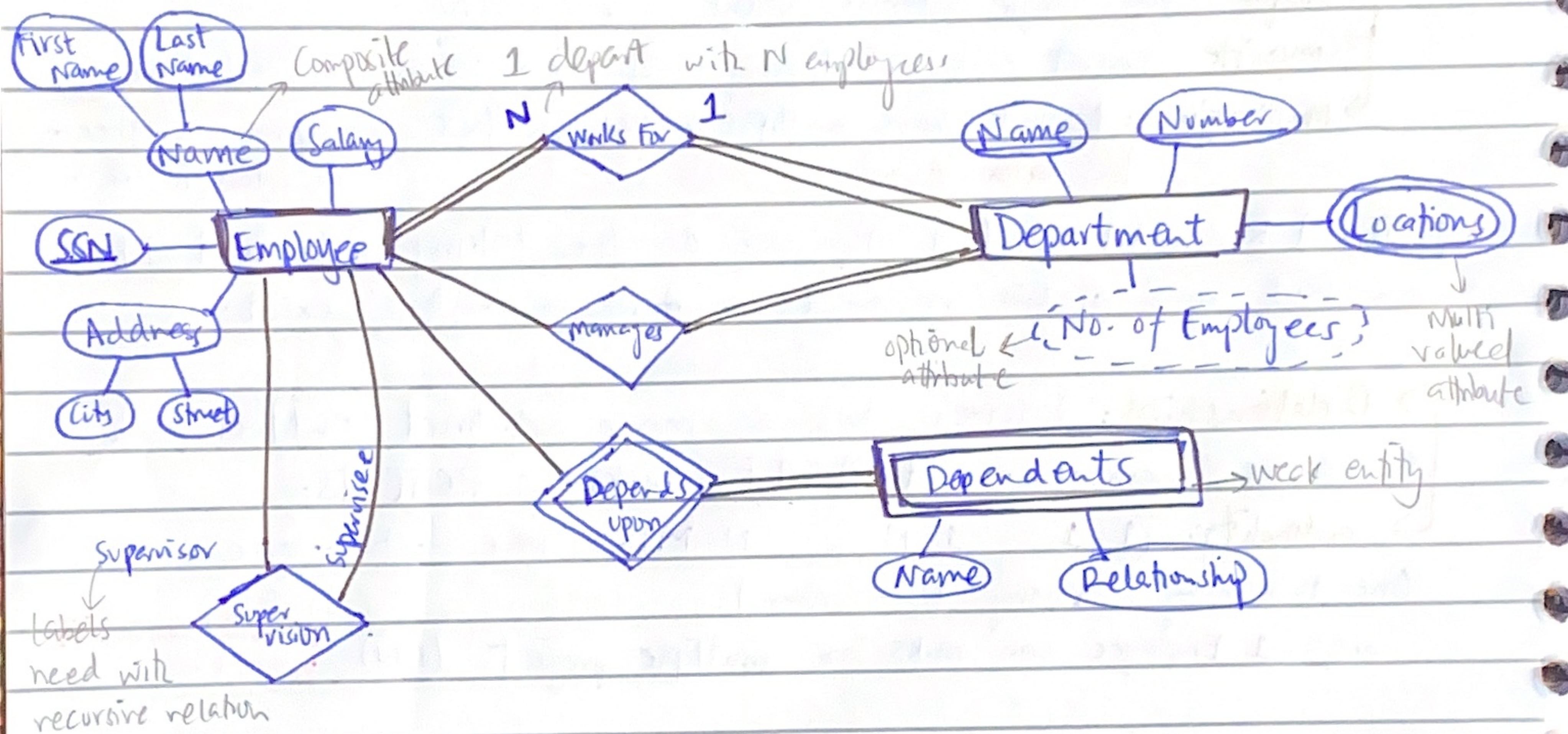
⇒ Recursive Relation: A self relationship e.g.: Employee can do with supervisor & supervisee. (labels are needed).

→ Desired attribute (optional): This can change on runtime based on state.
e.g.: No. of Employee. → Have dot circle around it.

→ Weak Entity: An entity does not have its own key attribute and it is in relation with an owner entity who has the key.
e.g.: 'Dependent' Entity is weak, EMPLOYEE is its owner.

→ The rectangle is double lined and diamond b/w both relations is also double lined.

→ Participation Constraint: A double line is used b/w entities on either or both side of relationship diamond to show which entities "must" and "not needed" for all of the instances of other entity.
 e.g: All departments have employees but all employees doesn't manages a depart.

⇒ Examples in Slides (Prac. Ques).